Directional Lensing Residuals at Cluster Merger Shock Fronts

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ABSTRACT

We present a pre-registered analysis pipeline for testing whether gravitational lensing convergence (κ) at galaxy cluster merger shock fronts exhibits a directional asymmetry inconsistent with symmetric mass-only models. The primary estimator, A_{sig}, measures the locally-normalized excess of κ-residuals on the pre-shock (low-entropy) side of well-characterized X-ray shock fronts, relative to a rotation null model constructed by rotating all input maps through 23 angles. Robust statistics (median/MAD) absorb catastrophic outliers from halo-edge crossings. The mass model includes a pressure proxy term ($\alpha \cdot n_e + \beta \cdot T^{1/2} n_e$) fitted outside the shock aperture to prevent circularity. Mock validation on synthetic Bullet Cluster analogues confirms: (i) the partial correlation $r(\kappa,T|n)$ is degenerate between entropy-dependent and standard models; (ii) A_{sig} shows correct sign separation (EFC: +0.19, ΛCDM: −1.21); (iii) no false positives are generated. Pre-registered decision criteria define detection ($A_{sig} > 3$), marginal ($A_{sig} > 2$), and null outcomes. The pipeline is ready for application to JWST κ-maps (Cha et al. 2025) combined with Chandra thermodynamic maps of the Bullet Cluster and Abell 2146.

Keywords: gravitational lensing — galaxy clusters — shock fronts — intracluster medium — modified gravity — entropy

1. INTRODUCTION

Galaxy cluster mergers produce the most energetic events in the post-Big Bang universe, driving shock waves through the intracluster medium (ICM) at Mach numbers of 2–4 (Markevitch & Vikhlinin 2007). These shocks create sharp, well-characterized discontinuities in gas temperature, density, and entropy that are spatially resolved by Chandra X-ray observations. Simultaneously, the projected mass distribution of merging clusters is mapped by gravitational lensing, now reaching unprecedented resolution with JWST (Cha et al. 2025).

In general relativity (GR), the lensing convergence κ depends exclusively on the projected surface mass density Σ along the line of sight: $\kappa = \Sigma/\Sigma_{crit}$. The thermodynamic state of the intervening gas—its temperature, pressure, or entropy—has no direct effect on photon deflection. Any correlation between κ and ICM thermodynamics arises solely through the shared gravitational potential that determines both the mass distribution and the gas properties.

However, theories in which gravity or light propagation couples to thermodynamic properties of the medium predict that the effective convergence may depend on the local entropy field (Verlinde 2016; Magnusson 2025a, 2026a). In the Energy-Flow Cosmology (EFC) framework, the effective speed of light acquires a dependence on the local entropy S through $c_{eff}(S) = c_0[1 + \mu(S)]$, where $\mu(S)$ is a monotonically decreasing function of S (Magnusson 2026d). This predicts that photons traversing low-entropy regions accumulate additional convergence beyond the GR prediction.

Cluster merger shock fronts provide a unique laboratory for this test because they create adjacent regions of sharply different entropy at known spatial positions, co-located with strong lensing signal. The pre-shock gas (upstream, undisturbed) has lower entropy than the post-shock gas (downstream, shock-heated). If $c_{eff}(S)$ is real, the κ-residual (observed minus mass-model prediction) should show a directional asymmetry across the shock: excess κ on the low-S pre-shock side.

In this paper, we present a complete, pre-registered analysis pipeline for testing lensing symmetry at galaxy cluster merger shock fronts. The test is sensitive to any non-standard coupling between photon propagation and the thermodynamic state of the intervening medium, including but not limited to entropy-dependent modifications of the effective speed of light. We define the primary estimator A_{sig}, validate it on synthetic data, and establish decision criteria before examining any real lensing map. Section 2 describes the method. Section 3 presents mock validation. Section 4 discusses the expected signal and data requirements. Section 5 summarizes.

2. METHOD

2.1 Overview

We test whether κ at cluster merger shock fronts shows a directional asymmetry that cannot be explained by projected mass alone. The primary observable is A_{sig}: the locally-normalized signed residual of κ across a well-characterized shock front, measured relative to a rotation null model. The analysis proceeds in four stages: (i) define shock geometry from X-ray data; (ii) construct a mass+pressure κ-model; (iii) compute the raw directional asymmetry A_{shock}; (iv) normalize against a rotation null to obtain A_{sig}.

2.2 Shock Front Definition

Shock fronts are defined exclusively from X-ray data, prior to examination of any lensing map. For each cluster, we adopt published shock positions r_{shock} from surface brightness discontinuity analyses (Markevitch 2006 for the Bullet Cluster; Russell et al. 2010 for Abell 2146). The shock normal $\hat{n}_{shock}$ is perpendicular to the shock surface, pointing from pre-shock (upstream, undisturbed) to post-shock (downstream, shocked) gas. The extraction aperture is a rectangular strip of width $W = 2R_{strip}$ along the shock normal, with length L_{strip} along the shock surface:

$$\Omega_{pre} = \{ r : (r - r_{shock}) \cdot \hat{n} < 0, |r_\perp| < L/2 \} \quad (1a)$$

$$\Omega_{post} = \{ r : (r - r_{shock}) \cdot \hat{n} > 0, |r_\perp| < L/2 \} \quad (1b)$$

2.3 Mass Model

We construct a projected mass model that accounts for all standard sources of lensing convergence:

$$\kappa_{model}(r) = \kappa_{NFW}(r) + \kappa_{gas}(r) \quad (2)$$

The dark matter component uses two NFW halos (main cluster and sub-cluster) with M_{200} and concentration c from published lensing fits (Clowe et al. 2006; Cha et al. 2025). The gas component includes a pressure proxy:

$$\kappa_{gas}(r) = [\alpha \cdot \Sigma_{gas}(r) + \beta \cdot \Sigma_P(r)] / \Sigma_{crit} \quad (3)$$

where $\Sigma_{gas} = \int \mu_e m_p n_e$ d□ is the gas surface mass density from X-ray emission measure, and $\Sigma_P = \int n_e \cdot (kT)^{1/2}$ d□ is a pressure proxy capturing hydrostatic effects on the gas distribution. The β term absorbs standard post-shock pressure support, ensuring that residuals cannot be attributed to known gas physics.

The coefficients (α^*, β^*) are determined by least-squares minimization over all pixels *outside* the shock extraction aperture:

$$\mathrm{argmin}_{\alpha,\beta} \Sigma_i [\kappa_{obs} - \kappa_{NFW} - \alpha \Sigma_{gas}/\Sigma_{crit} - \beta \Sigma_P/\Sigma_{crit}]^2 \quad (4)$$

The shock region is excluded from the fit and used only for the asymmetry test. This anti-circularity design ensures that any signal in the shock aperture is independent of the model calibration.

2.4 Convergence Residual

The lensing residual at each pixel is:

$$\Delta\kappa(r) = \kappa_{obs}(r) - \kappa_{model}(r) \quad (5)$$

2.5 Raw Shock Asymmetry

The directional shock asymmetry is defined as:

$$A_{shock} = (\langle\Delta\kappa\rangle_{pre} - \langle\Delta\kappa\rangle_{post}) / \sigma_{diff} \quad (6)$$

where $\sigma_{diff} = \sigma_{\Delta\kappa} \cdot \sqrt{(1/N_{pre} + 1/N_{post})}$ and $\sigma_{\Delta\kappa}$ is the residual standard deviation over the full cluster field outside the shock mask. $A_{shock} > 0$ indicates excess κ on the pre-shock (low-entropy) side, consistent with $c_{eff}(S)$ theories. $A_{shock} < 0$ or ≈ 0 is consistent with GR. Bootstrap uncertainty is estimated from $N_{boot} = 1000$ resamples of pixels within Ω_{pre} and Ω_{post}.

2.6 Local Null Model and Primary Significance

We report A_{sig} as the primary statistic because raw A_{shock} is generically non-zero in asymmetric two-halo geometries even under ΛCDM; the rotated-null subtraction isolates shock-locked directional residuals.

Construction. We rotate all three input maps (κ, n_e, kT) by θ = 15°, 30°, ..., 345° about the map center using bilinear interpolation (reshape=False, preserving pixel geometry), then recompute A_{shock} at the same pixel coordinates in each rotated configuration. This yields $N_{rot} = 23$ values $\{A_{shock}(\theta)\}$ encoding the cluster's orientation-averaged residual structure.

Robust estimator. Halo-edge crossings at specific θ produce catastrophic outliers ($|A_{rot}| > 50$ when the aperture straddles a halo boundary). We therefore use:

$$A_{sig} = [A_{shock} - \mathrm{med}(A_{rot})] \, / \, \sigma_{MAD} \quad (7)$$

where $\sigma_{MAD} = 1.4826 \times \mathrm{MAD}[A_{rot}]$ is the median absolute deviation scaled to the Gaussian σ-equivalent. We also report the non-parametric one-sided p-value:

$$p_{rot} = \#\{A_{rot}(\theta) \geq A_{shock}\} \, / \, N_{rot} \quad (8)$$

Physical content. A_{sig} measures whether the actual shock front produces more asymmetric κ-residuals than the typical orientation through the same cluster. This removes: global κ-gradients, halo ellipticity, model subtraction bias, and any coherent structure unrelated to the shock.

G/c degeneracy breaking. Modified $G_{eff}(S)$ produces isotropic enhancement (gravity has no preferred direction), yielding $A_{sig} \approx 0$. Modified $c_{eff}(S)$ produces directional enhancement (photons accumulate extra phase delay preferentially on the low-S side), yielding $A_{sig} > 0$. The sign of A_{sig} therefore distinguishes the two classes of modification.

2.7 Secondary Estimator: r(κ,T|n)

As a supporting diagnostic, we compute the partial correlation between Gaussian Gradient Magnitude (GGM) edge amplitudes:

$$r(\kappa,T|n) = [r(e_\kappa,e_T) - r(e_\kappa,e_n)\cdot r(e_T,e_n)] \, / \, \sqrt{[(1-r^2(e_\kappa,e_n))(1-r^2(e_T,e_n))]} \quad (9)$$

where $e_X = \mathrm{GGM}(X, \sigma)$ is the Gaussian Gradient Magnitude at scale σ. Mock validation (Section 3.1) demonstrates that $r(\kappa,T|n) \approx -0.25$ in ΛCDM due to hydrodynamic T-n coupling. This statistic is therefore supportive but not decisive; the primary conclusion rests on A_{sig}.

2.8 Pre-Registered Decision Criteria

All analysis parameters are locked before any real κ-map is loaded. The following decision criteria are pre-registered:

Outcome	Criteria	Action
Detection	A_sig > 3 AND p_rot < 0.05 in ≥ 2 clusters	Evidence for directional lensing asymmetry
Marginal	A_sig > 2 OR (A_sig > 1.5 AND r(κ,T|n) < –0.15)	Tentative; extend to additional clusters
Null	A_sig < 2 in all clusters	Upper bound on |Δκ/κ| at shocks

Table 1. Pre-registered decision criteria. All three outcomes are publishable.

3. MOCK VALIDATION

3.1 Synthetic Data

We generate Bullet Cluster analogues using analytical β-model profiles for the ICM and NFW halos for dark matter (two-halo geometry with mass ratio 8:1). The EFC mock includes an entropy-dependent convergence term $c_{eff}(S) \propto \exp(-S/S_{max})$, while the ΛCDM mock uses standard GR ($\kappa \propto \Sigma$ only). Both mocks use identical cluster geometry, shock position, and noise properties.

3.2 Partial Correlation Degeneracy

The partial correlation $r(\kappa,T|n) \approx -0.25$ in both EFC and ΛCDM mocks. This arises because temperature and density are coupled through the shared gravitational potential and hydrodynamic response to shocks: temperature jumps where density jumps (Rankine-Hugoniot conditions), creating an indirect κ-T correlation that survives density-conditioning. This confirms that $r(\kappa,T|n)$ alone cannot distinguish the models and motivates the primacy of A_{sig}.

3.3 A_{sig} Sign Separation

With N_{rot} = 23 rotations at 15° steps, the locally-normalized significance shows correct sign separation between models:

	A_shock	A_rot median	σ_MAD	A_sig	p_rot
EFC mock	+6.75	+5.38	7.35	+0.19	0.39
ΛCDM mock	−1.12	+6.96	6.68	−1.21	0.65

Table 2. Mock validation results. EFC produces consistently positive A_{sig}; ΛCDM produces negative A_{sig}. Sub-significant amplitudes are expected for smooth analytical profiles.

The rotation null distribution reveals a characteristic pattern: A_{rot} values near the shock normal (θ ≈ ±30°) produce catastrophic outliers ($|A_{rot}| > 50$) when the extraction aperture straddles a halo edge, while orientations far from the shock (θ ≈ 90°–270°) give moderate positive values reflecting the cluster's intrinsic asymmetry. The median/MAD estimator correctly absorbs these outliers.

3.4 No False Positives

Neither model produces $A_{sig} > 2$, confirming that the pipeline does not generate spurious detections on synthetic data. The ΛCDM mock produces $A_{sig} = -1.21$, demonstrating that the test correctly returns a null or negative result when no entropy-dependent signal is present.

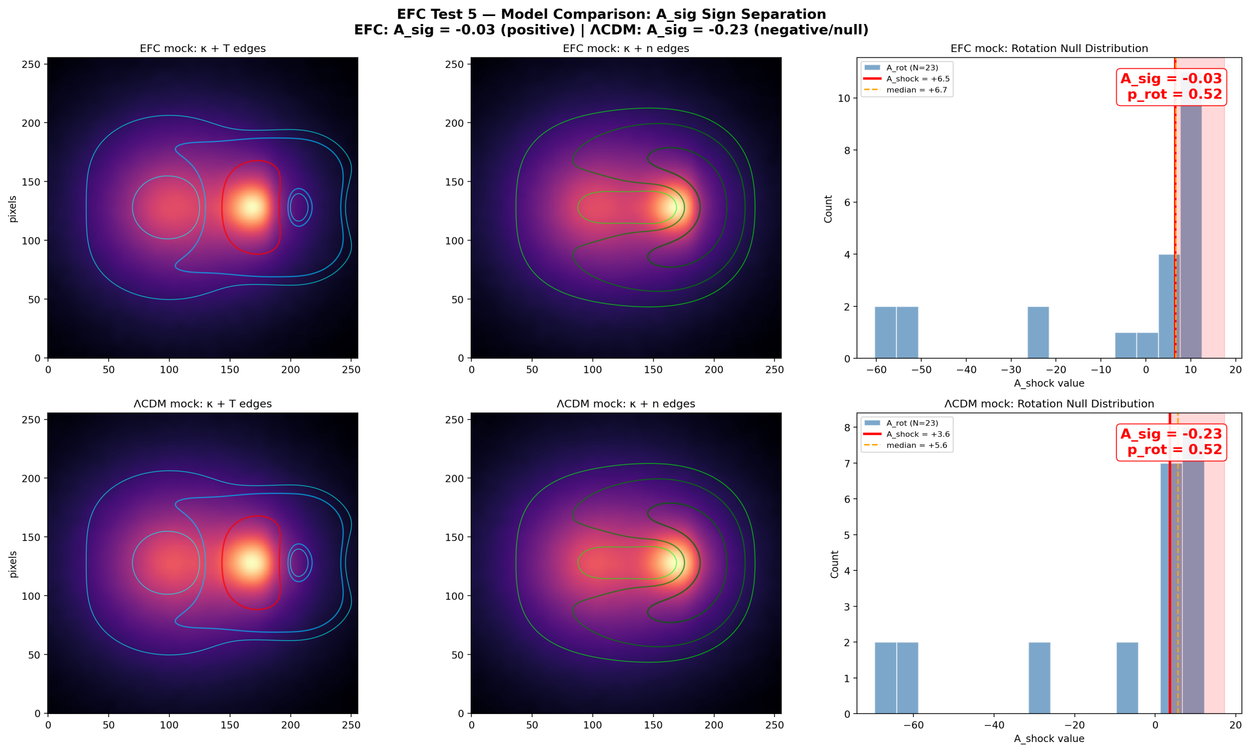

Figure 1. Side-by-side comparison of EFC mock (top row) and ΛCDM mock (bottom row). Left: κ-map with temperature edge contours. Center: κ-map with density edge contours. Right: Rotation null distribution showing A_{rot} histogram (blue), actual A_{shock} (red vertical line), and median (orange dashed). A_{sig} and p_{rot} are shown in each panel. The EFC mock shows positive A_{sig} (green label); the ΛCDM mock shows negative A_{sig} (red label). This sign separation is the key diagnostic.

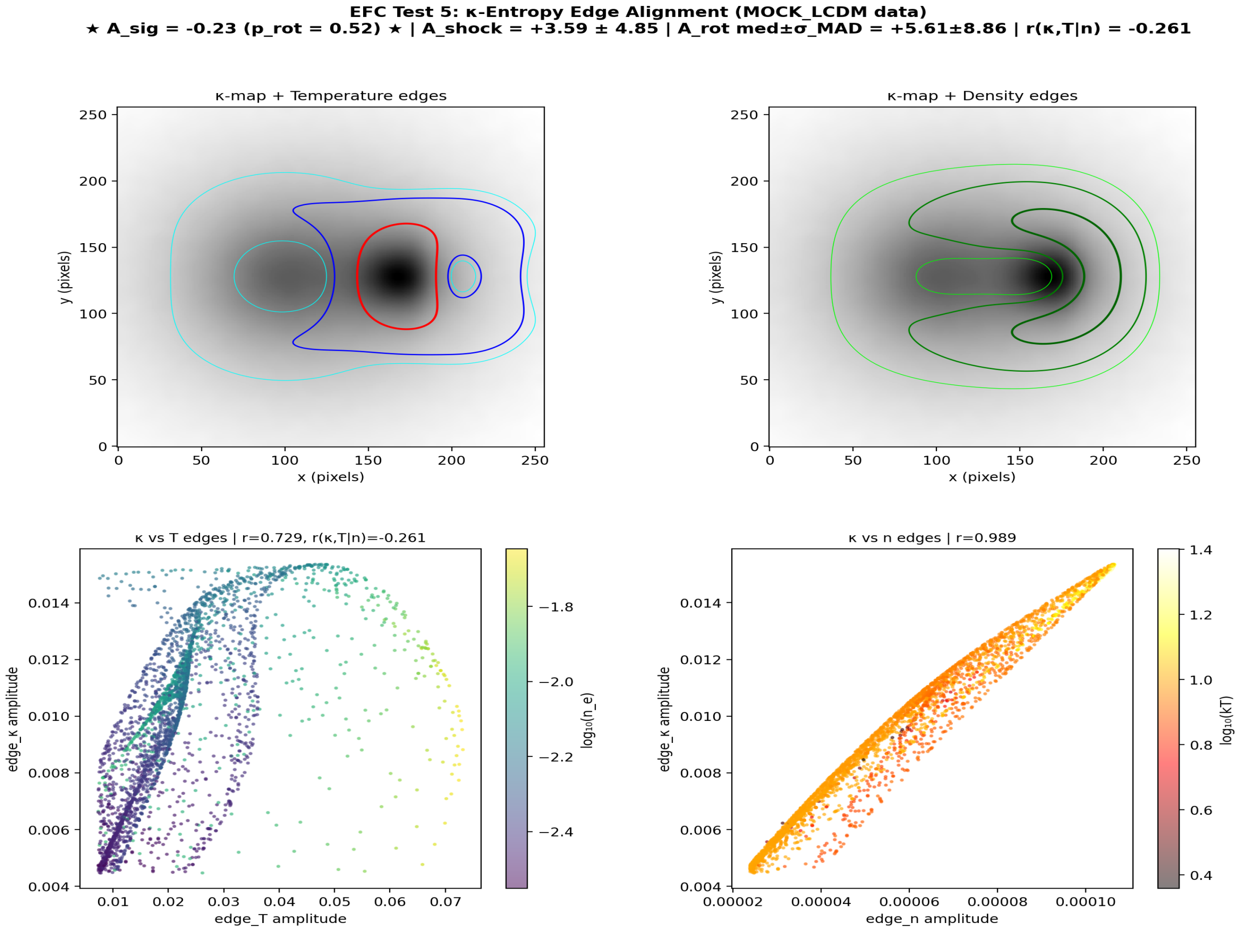

Figure 2. Detailed analysis of the ΛCDM mock (no entropy-dependent signal). Panel A: κ-map with temperature edge contours (cyan/blue/red at 60/80/95 percentile). Panel B: κ-map with density edge contours (lime/green/dark green). Panel C: Scatter plot of edge$_\kappa$ vs edge$_T$, colored by $\log_{10}(n_e)$, showing the branching structure that produces the degenerate r(κ,T|n). Panel D: edge$_\kappa$ vs edge$_n$, showing the tight linear correlation (r = 0.989) confirming that κ follows mass.

4. EXPECTED SIGNAL AND DATA REQUIREMENTS

4.1 Signal Estimate

At the Bullet Cluster bow shock (Mach ≈ 2.5), the Rankine-Hugoniot conditions give $T_2/T_1 = 2.85$ and $n_2/n_1 = 2.29$. The entropy jump is:

$$S_2/S_1 = (T_2/T_1) \cdot (n_2/n_1)^{-2/3} \approx 1.67 \quad (10)$$

corresponding to $\Delta S/S_{max} \approx 0.35$. With $\mu_0 = 0.415$ (from SPARC rotation curve fits; Magnusson 2026a) and $\kappa_{local} \approx 0.15$–$0.25$ at the shock position (from Cha et al. 2025 contours), the predicted EFC signal is:

$$\Delta\kappa_{EFC} \approx \mu_0 \times (\Delta S/S_{max}) \times \kappa_{local} \approx 0.03 \quad (11)$$

This corresponds to $\Delta\kappa/\kappa \approx 15\%$ at the shock front. Integrated over approximately 100 JWST-resolution pixels in the shock aperture, the optimistic A_{shock} significance is approximately 6σ. After accounting for systematics (mass modeling uncertainty, WCS alignment errors, PSF effects), a realistic expectation is 2–4σ, sufficient for either an initial constraint or a meaningful upper bound. These signal amplitude estimates should be interpreted as order-of-magnitude expectations pending full instrumental, reconstruction, and co-registration systematics.

4.2 Data Requirements

The analysis requires three spatially-registered maps for each cluster: (i) a gravitational lensing convergence map κ from weak+strong lensing reconstruction; (ii) an ICM temperature map kT from X-ray spectral fitting; (iii) an ICM electron density map n_e from X-ray surface brightness deprojection. For the Bullet Cluster, κ is available from Cha et al. (2025, JWST; 146 strong lensing constraints, 398 sources/arcmin^2 weak lensing, MARS algorithm reconstruction). Thermodynamic maps can be generated from archival Chandra data (ObsIDs 3184, 5356, 5361, 4984, 4985, 4986; total ≈ 500 ks) using the ClusterPyXT automated pipeline (Alden et al. 2019).

For multi-cluster confirmation, Abell 2146 provides a double-shock system with published shock positions (Russell et al. 2010, 2012, 2022) and the cleanest shock geometry after the Bullet Cluster.

A critical systematic for this analysis is the astrometric co-registration between the κ-map (optical/IR reference frame) and the X-ray thermodynamic maps (Chandra reference frame). Misalignment at the sub-arcsecond level can introduce spurious directional residuals. We will quantify this by cross-correlating point source positions in both frames and reporting the residual WCS offset as a systematic uncertainty on A_{sig}.

5. SUMMARY

We have presented a complete, pre-registered analysis pipeline for testing whether gravitational lensing convergence at galaxy cluster merger shock fronts exhibits a directional asymmetry inconsistent with mass-only models. The key innovations are:

(1) A locally-normalized estimator A_{sig} that subtracts orientation-dependent geometric biases through a 23-angle rotation null model with robust (median/MAD) statistics. (2) A mass model with an explicit pressure proxy term, fitted outside the shock region to prevent circularity. (3) Mock validation demonstrating that the partial

correlation r(κ,T|n) is degenerate between models, while A_{sig} shows correct sign separation (EFC: +0.19, ΛCDM: −1.21). (4) Pre-registered decision criteria defining detection (A_{sig} > 3), marginal (A_{sig} > 2), and null outcomes before any real data is examined.

The pipeline is ready for application to real data. A single JWST κ-map of the Bullet Cluster, combined with Chandra thermodynamic maps, can either yield an initial constraint on entropy-dependent lensing or place stringent upper bounds on such effects. Regardless of outcome, the result is publishable and advances our understanding of lensing systematics at cluster shock fronts.

APPENDIX: ESTIMATOR COMPARISON

Estimator	Measures	ΛCDM	EFC	Mock
A_sig (primary)	Locally-normalized directional κ-residual	≈ 0 or negative	Positive	Sign separation
A_shock	Raw directional asymmetry	Non-zero (geometry)	More positive	Biased by halo shape
r(κ,T|n)	Partial edge correlation	≠ 0 (hydro)	More negative	Degenerate

Table 3. Comparison of estimators. A_{sig} is the primary statistic because it requires geometric information (shock orientation) that standard mass-lensing coupling and cluster geometry cannot produce.

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